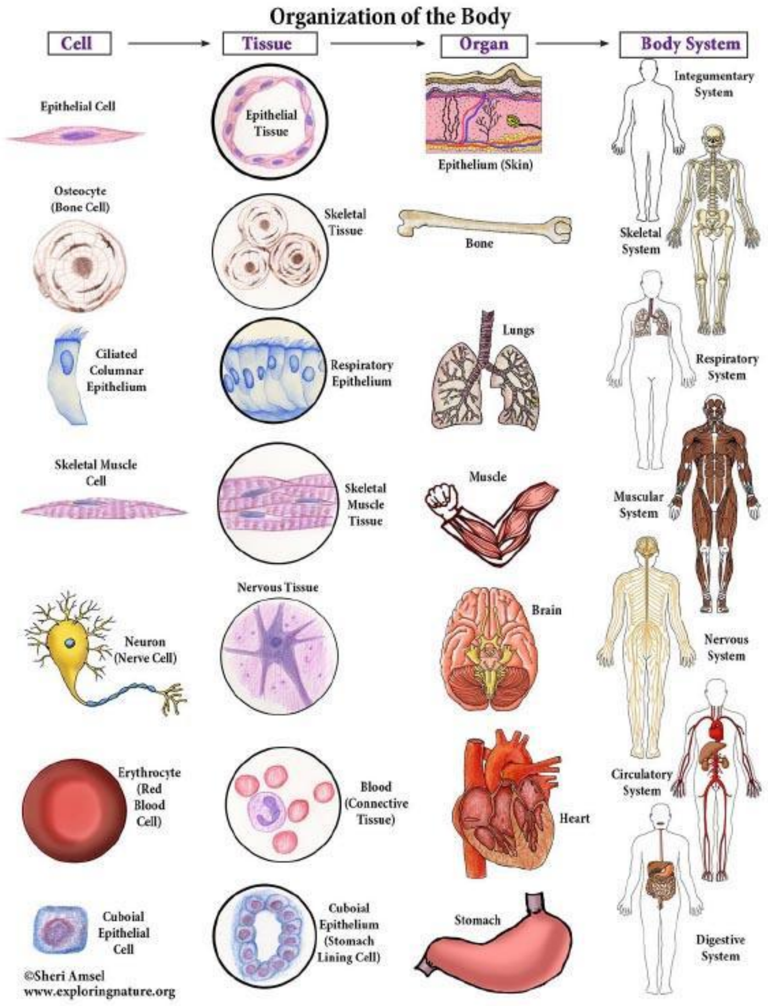
# Human body: Tissue, Organ and Glands

Course: BIO 101
Introduction to Biology

Chapter: 5

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# Level of organization

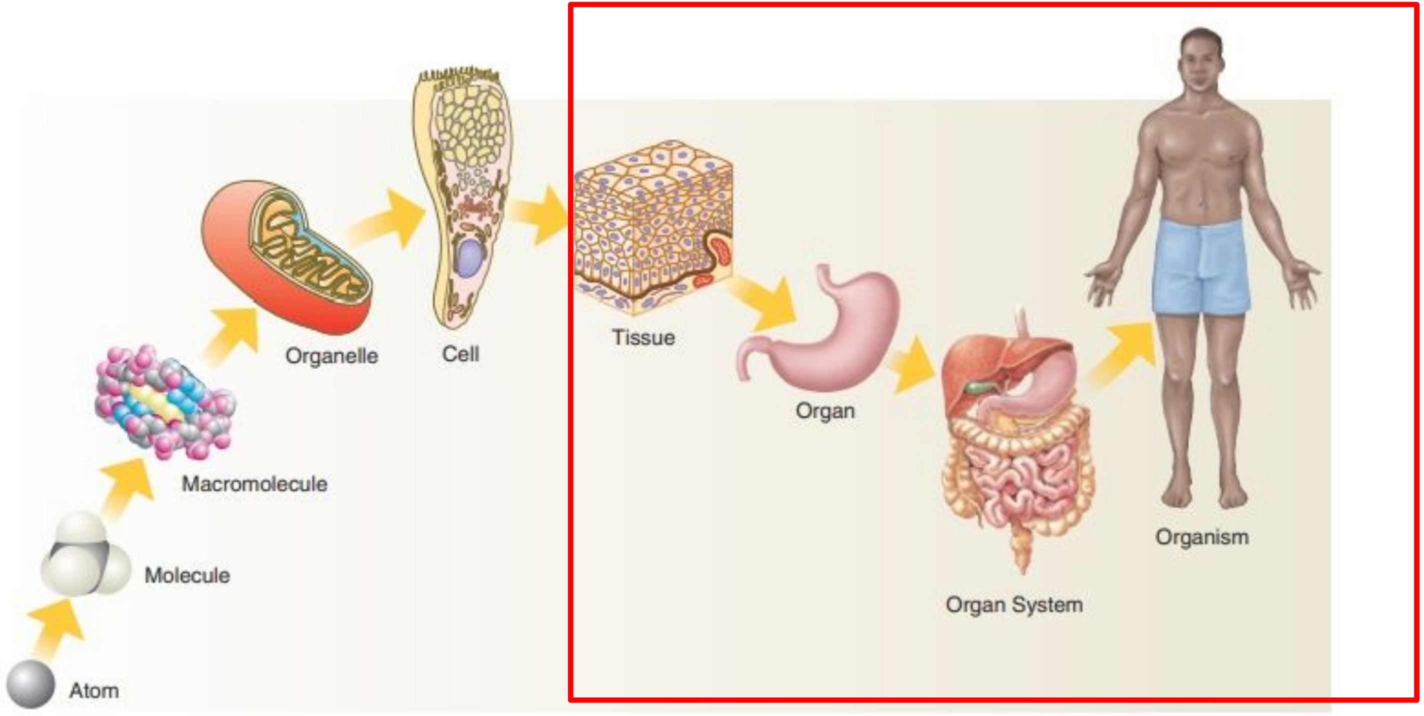



FIGURE 1-1 Organization levels of the body.

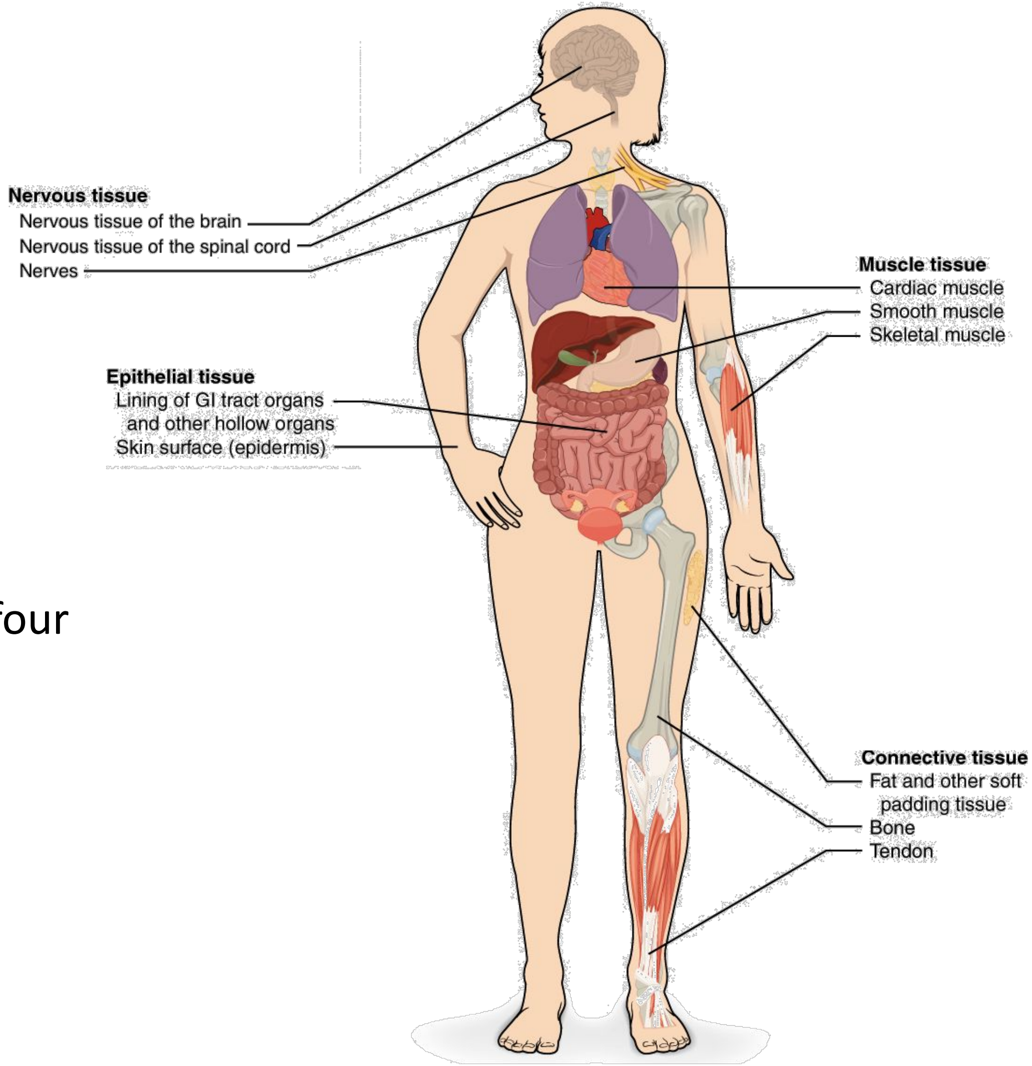
Adapted from Shier, D.N., Butler, J.L., and Lewis, R. Hole's Essentials of Human Anatomy & Physiology, Tenth edition. McGraw Hill Higher Education, 2009.

# Tissue

- Group of cells having similar structure
- Usually found together
- Perform relatively common function
- Held together by extracellular fluid and fibers

More than 200 cell types make up four major tissues of the body.

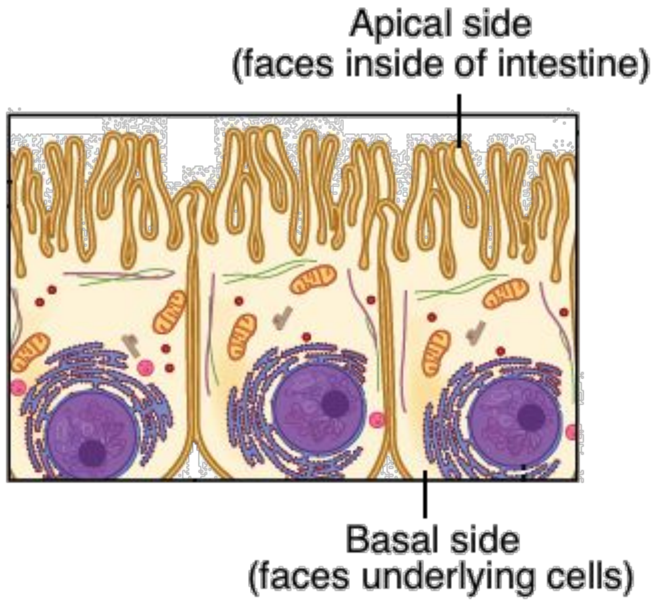
1. Epithelial tissue
2. Connective tissue
3. Muscle tissue
4. Nervous tissue



# Epithelial tissue

- Consists of tightly packed cells arranged in a continuous sheet, in one or more layers.
- Contains little extracellular material

- Makes up the epidermis of the skin
- Covers the surface of organs
- Lines cavities, forms tubes and ducts
- Provides the secreting portions of glands
- As tightly packed, helps them act as barriers to the movement of fluids and potentially harmful microbes
- e.g.
  - Skin
  - Internal covering of GIT

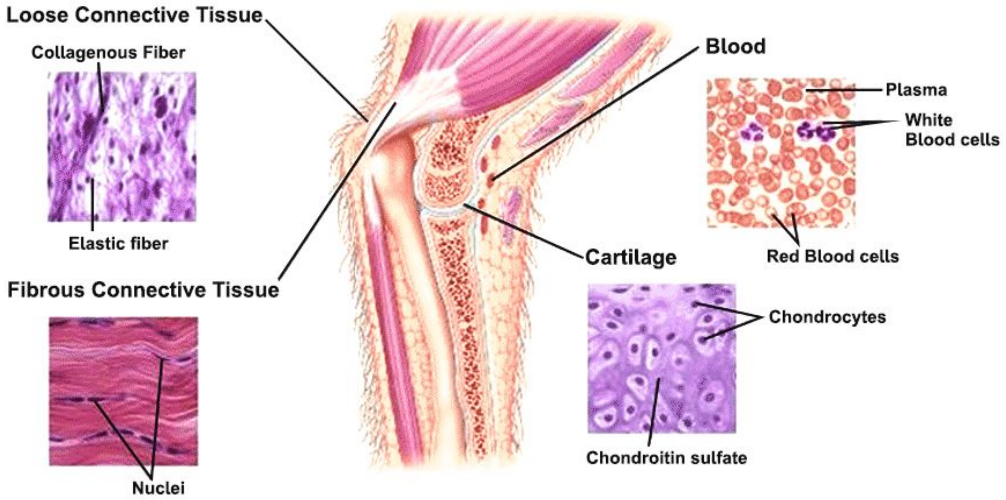


# Connective tissue

- Most abundant tissue in the body
- Contains abundant extracellular material

- Made up of diverse cell types including fibroblasts, fat and blood cells

- Supports and connects other tissues and organs



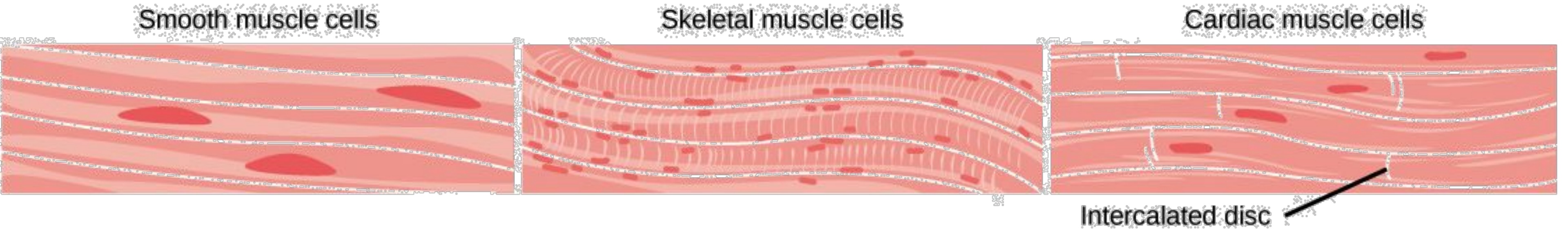
# Muscle tissue

- Essential for keeping the body upright, allowing it to move, and even pumping blood and pushing food through the digestive tract.

- Muscle cells, often called muscle fibers
- Cells that are able to contract

- Contain contractile protein myosin and actin filaments

- E.g.
    - Skeletal(voluntary)
    - Smooth(involuntary)
    - Cardiac(involuntary)



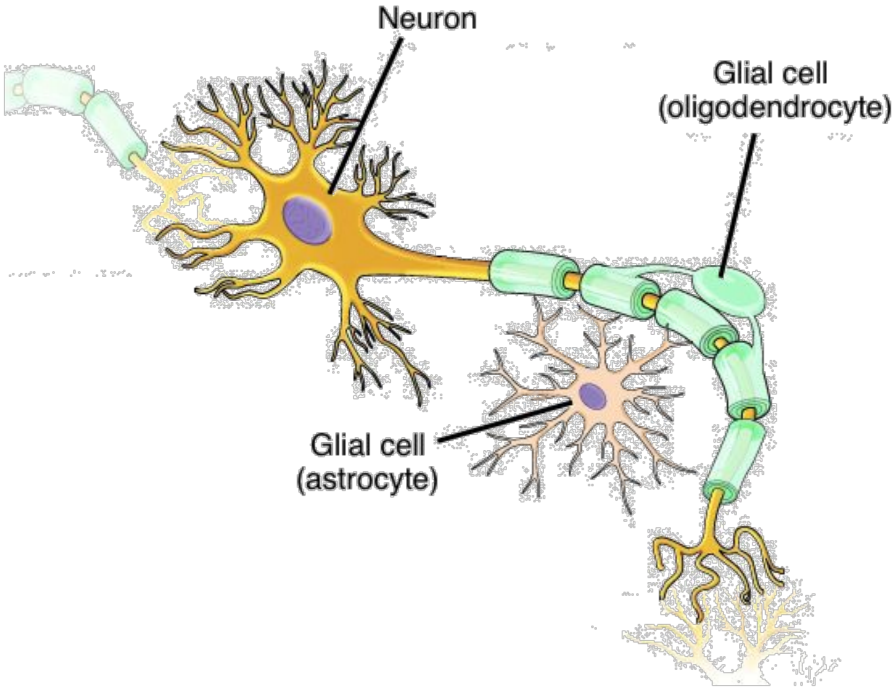
# Nervous tissue

- Highly specialized tissue
- Can be stimulated by sensing a stimuli -> transmit nerve impulse from one to another part of the body

- Allows to move, think, taste etc.
- Create an awareness of the environment

- Consists of two main types of cells: neurons/ nerve cells and glial cells

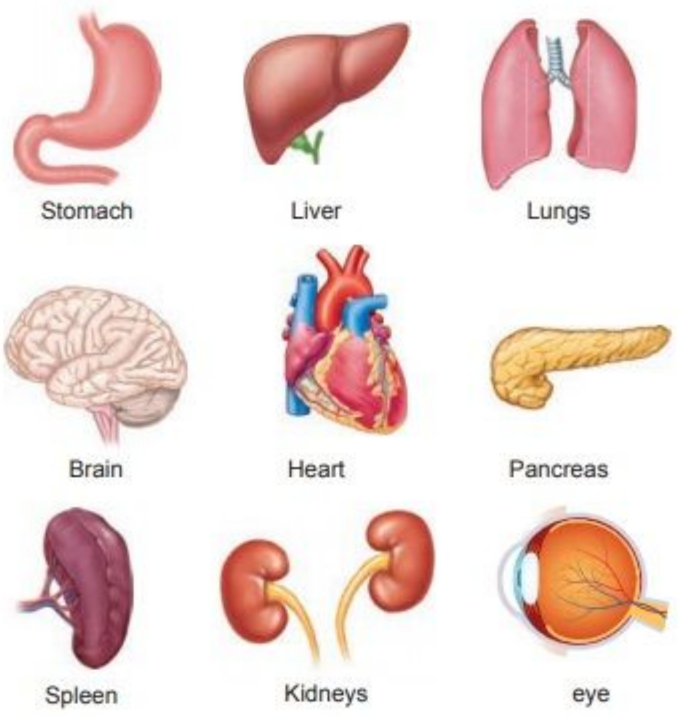
  - **neurons** are the basic functional unit of the nervous system
  - **glial** mainly act to support neuronal function.



# Organ

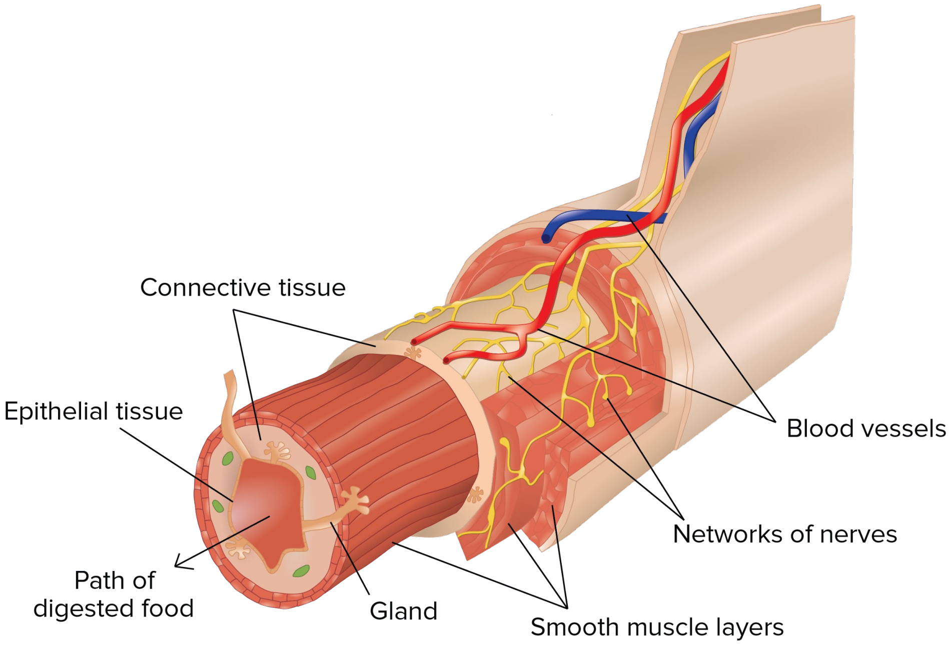
- Collection of tissues united to perform a particular function.

- E.g.
- Heart is an organ that pumps blood through body
- Heart consists of muscle tissue (cardiac muscle), nervous tissue, connective tissue (blood, adipose) , and epithelial tissue (epicardium)



# Organ

- Most organs contain **all four types of tissue**
- The inside of the intestine is lined by **epithelial cells**
- Around the epithelial layer are layers of **connective tissue** and smooth muscle, interspersed with glands, blood vessels, and neurons
- **smooth muscle** contracts to move food through the gut, under the control of its associated networks of **neurons**

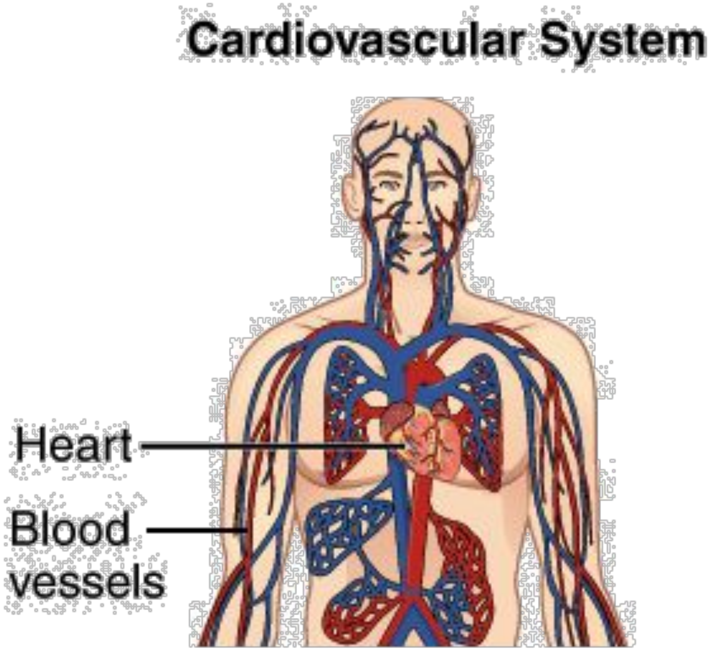



Gastrointestinal tract

# Organ systems

- Organs are grouped into organ systems
- They work together to carry out a particular function for the organism

- The heart and the blood vessels make up the cardiovascular system
- Work together to circulate the blood, bringing oxygen and nutrients to cells throughout the body and carrying away carbon dioxide and metabolic wastes.



# Major organ systems of the human body

| Organ system | Function | Organs, tissues, and structures involved |
|---|---|---|
| Cardiovascular | Transports oxygen, nutrients, and other substances to the cells and transports wastes, carbon dioxide, and other substances away from the cells; it can also help stabilize body temperature and pH | Heart, blood, and blood vessels |
| Lymphatic | Defends against infection and disease and transfers lymph between tissues and the blood stream | Lymph, lymph nodes, and lymph vessels |

# Major organ systems of the human body

| | | |
|---|---|---|
| Digestive | Processes foods and absorbs nutrients, minerals, vitamins, and water | Mouth, salivary glands, esophagus, stomach, liver, gallbladder, exocrine pancreas, small intestine, and large intestine |
| Endocrine | Provides communication within the body via hormones and directs long-term change in other organ systems to maintain homeostasis | Pituitary, pineal, thyroid, parathyroids, endocrine pancreas, adrenals, testes, and ovaries. |

# Major organ systems of the human body

| | | |
|---|---|---|
| Integumentary | Provides protection from injury and fluid loss and provides physical defense against infection by microorganisms; involved in temperature control | Skin, hair, and nails |
| Muscular | Provides movement, support, and heat production | Skeletal, cardiac, and smooth muscles |
| Nervous | Collects, transfers, and processes information and directs short-term change in other organ systems | Brain, spinal cord, nerves, and sensory organs—eyes, ears, tongue, skin, and nose |

# Major organ systems of the human body

| | | |
|---|---|---|
| Reproductive | Produces gametes—sex cells—and sex hormones; ultimately produces offspring | Fallopian tubes, uterus, vagina, ovaries, mammary glands (female), testes, vas deferens, seminal vesicles, prostate, and penis (male) |
| Respiratory | Delivers air to sites where gas exchange can occur | Mouth, nose, pharynx, larynx, trachea, bronchi, lungs, and diaphragm |

# Major organ systems of the human body

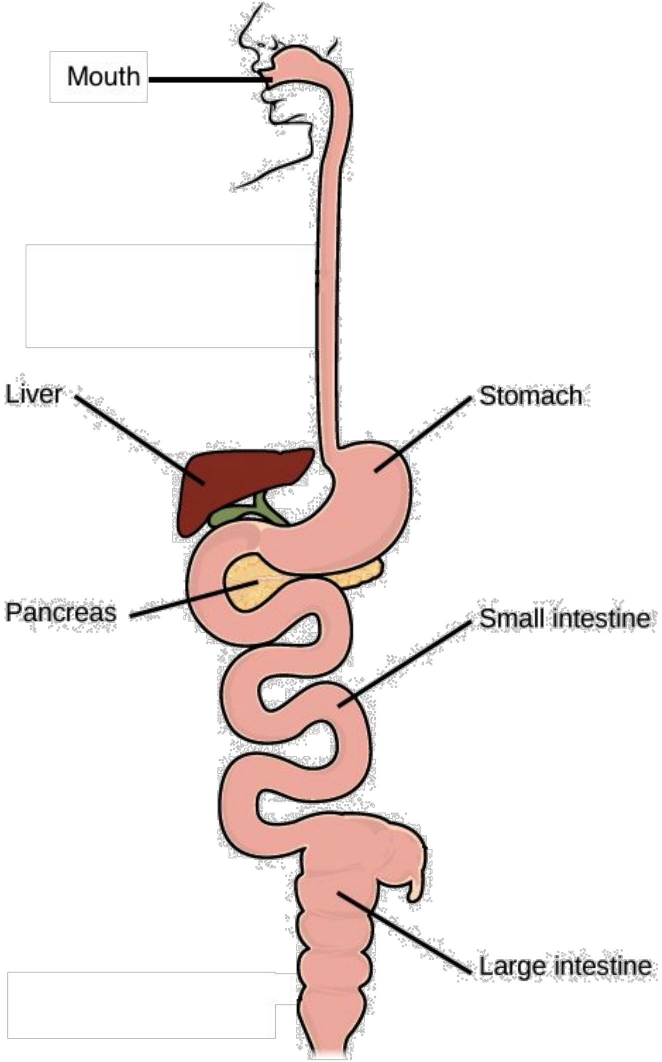
| | | |
|---|---|---|
| Skeletal | Supports and protects soft tissues of the body; provides movement at joints; produces blood cells; and stores minerals | Bones, cartilage, joints, tendons, and ligaments |
| Urinary | Removes excess water, salts, and waste products from the blood and body and controls pH | Kidneys, ureters, urinary bladder, and urethra |
| Immune | Defends against microbial pathogens—disease-causing agents—and other diseases | Leukocytes, tonsils, adenoids, thymus, and spleen |

# Organs in a system work together

- Organ system must work together for the system to function as a whole.

- The mouth, stomach, small intestine, and other digestive system organs work together to digest food and to absorb nutrients efficiently.

- Digestion wouldn't work well if your stomach stopped churning or if one of your enzyme-producing glands—like the pancreas—decided to take the day off!



# Organ systems work together as well!

- Different organ systems also cooperate to keep the body running.
- For example, the **respiratory system** and the **circulatory system** work closely together to deliver oxygen to cells and to get rid of the carbon dioxide the cells produce.

- The circulatory system picks up oxygen in the lungs and drops it off in the tissues, then performs the reverse service for carbon dioxide.
- The lungs expel the carbon dioxide and bring in new oxygen-containing air.
- Only when both systems are working together can oxygen and carbon dioxide be successfully exchanged between cells and the environment.

- The blood in the **circulatory system** has to receive nutrients from **digestive system before delivering them to the rest of the body.** Blood also undergoes filtration in kidneys (urinary/ excretory system) to remove the wastes they produce.

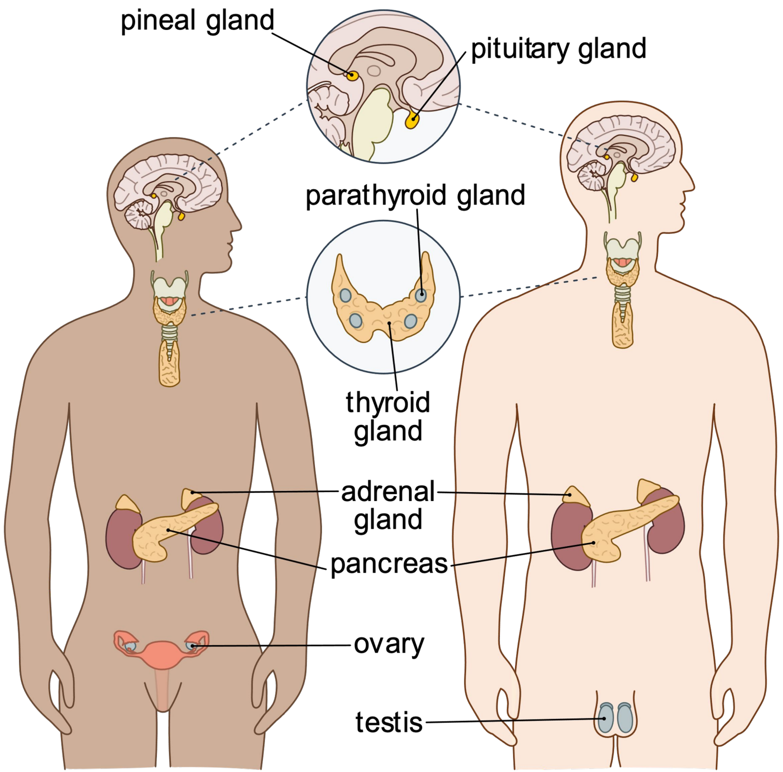
# Control and coordination

- Body functions are controlled by the nervous system and also by the endocrine system.
- Use **chemical messengers** to affect the function of other organ systems and to coordinate activity at different locations in the body.

- The nervous system uses **neurotransmitters** that carry **electrical impulses** to collect, process, and respond to information about the environment.
- The endocrine system produces and uses **chemical signals** called **hormones**, which travel through the bloodstream and control the actions of cells and organs.

# Gland

- Glands are organs that produce and release substances that perform certain functions.
- Located throughout the body
- 
- Two types:
    - Endocrine and
    - Exocrine



# Endocrine glands

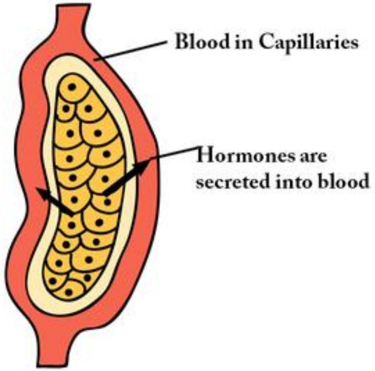
- Part of the endocrine system
- **Ductless glands**
- Make **hormones** and release them **directly into the bloodstream**
- Control important functions in the body:
  - growth and development
  - metabolism
  - mood
  - reproduction

**Examples of glands**

- adrenal glands
- pituitary gland
- hypothalamus
- thyroid
- pineal gland

There are also organs that contain endocrine tissue: epithelial cells create and secrete hormones. These include:

- pancreas
- kidneys
- ovaries
- testes



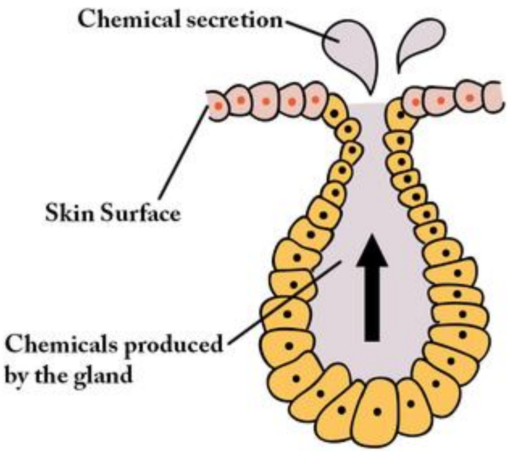

Endocrine Gland

# Exocrine glands

- Produce usually substances that are **not hormones**
- **Ducts present**
- Release substances through ducts to the **exterior of the body**
- Functions:
  - regulate body temperature
  - protect skin and eyes
  - produce breast milk

### Examples of glands

- salivary
- sweat
- mammary
- sebaceous
- lacrimal




Exocrine Gland

# Common glands and hormones

| Hormone | Gland produced in | Role |
| --- | --- | --- |
| Thyroid hormone | Thyroid | Regulates metabolism |
| Adrenaline (epinephrine) | Adrenal gland | Involved in "fight or flight" response |
| Cortisol | Adrenal gland | Involved in "fight or flight" response, regulates metabolism and immune responses |

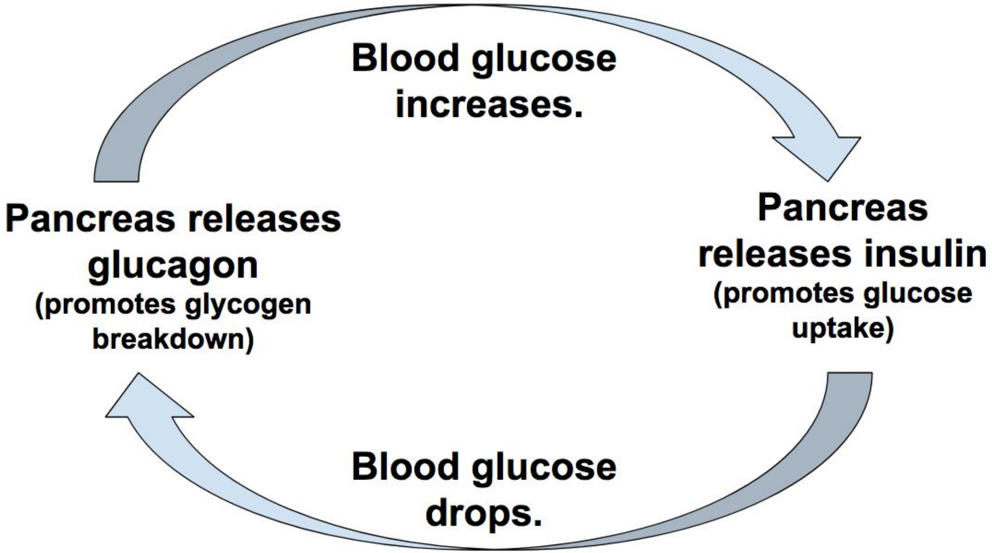
# Common glands and hormones

| Hormone | Gland produced in | Role |
| --- | --- | --- |
| Estrogen | Ovaries | Sexual and reproductive development, mainly in women |
| Testosterone | Testes, sometimes adrenal glands or ovaries | Sexual and reproductive development, mainly in men |
| Insulin | Pancreas | Blood sugar regulation, fat storage |
| Glucagon | Pancreas | Blood sugar regulation |

# Who regulates THE REGULATORS?

- The endocrine system is regulated by **negative feedback** mechanism to maintain **homeostasis.**
- Negative feedback mechanism: oppose a change -> bringing it back to set point

- For example, blood glucose regulation.



Thank You,
but sad to see
you go!